

CLOUD COMPUTING LAB 2 REPORT

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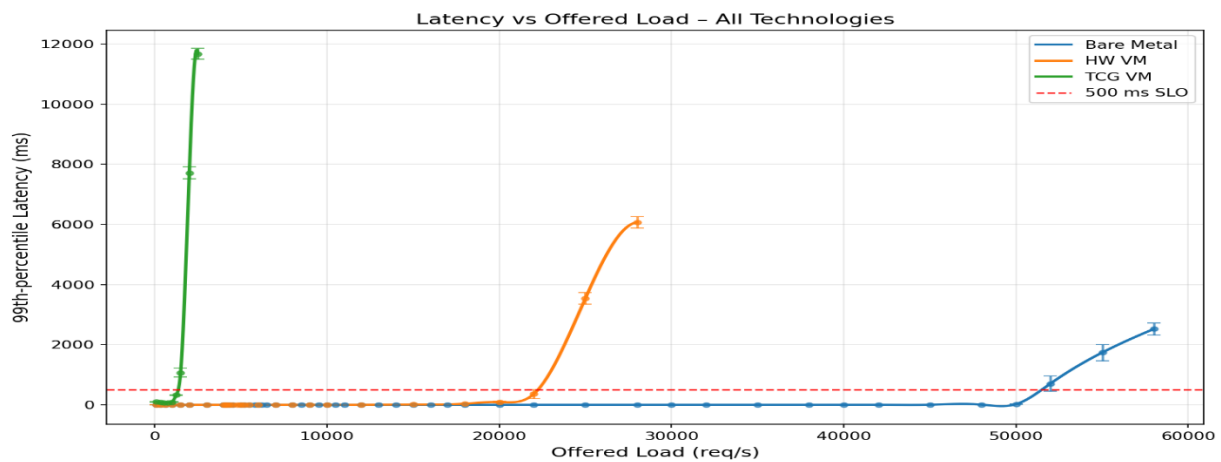
Questions

- 1) What is the bandwidth between the client (`client0`) and the server machine (`server0`) on bare metal? What is the bandwidth between the client (`client0`) machine and `vm1`? Are they different? Why? (5 Marks)

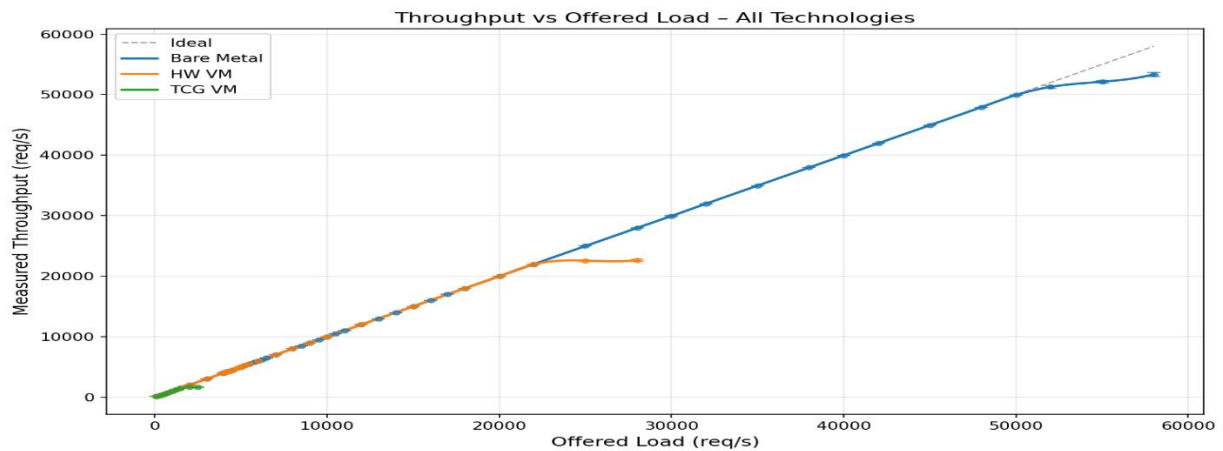
The measured bandwidth between the client (client0) and the server (server0) on bare metal was approximately **9.40 Gbps**, while the bandwidth between the client and the virtual machine (vm1) was slightly lower at **9.18 Gbps**.

Reason: The slight difference observed is due to virtualization overhead in the VM environment. In the case of bare metal, the network traffic is handled directly by the physical hardware, allowing for optimal performance. In contrast, communication with the virtual machine involves additional layers, such as the hypervisor and virtual network interfaces, which introduce minor processing overhead. As a result, although virtual machines can achieve near-native performance, bare metal systems still provide marginally higher bandwidth due to their direct access to hardware resources.

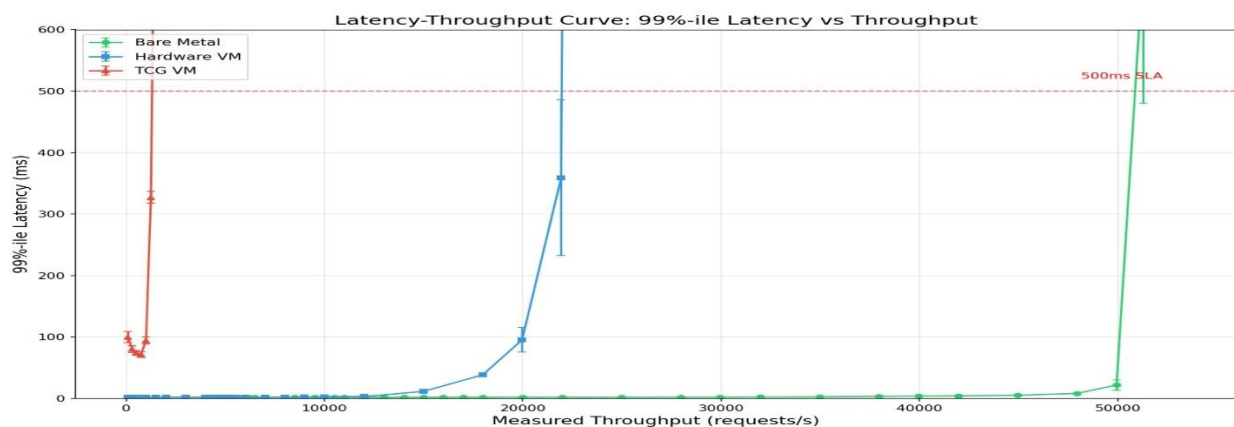
- 2) Draw the following graphs (include the error bars for each data point, and use different colors or markers to distinguish between the different technologies):
- i) Latency-Load curve for each of the three virtualization technologies. The x-axis should be the offered load (requests/s) and the y-axis should be the 99%-ile latency (ms). (5 Marks)



- ii) **Throughput-Load curve for each of the three virtualization technologies. The x-axis should be the offered load (requests/s) and the y-axis should be the measured throughput (requests/s). (5 Marks)**



- iii) **Latency-Throughput curve for each of the three virtualization technologies. The x-axis should be the measured throughput (requests/s) and the y-axis should be the 99%-ile latency (ms). (5 Marks)**



b) You should be able to interpret the graphs, and additionally answer the following questions: What are the characteristics of each curve? For graph 3, where is the max point¹ and what does it signify? (5 Marks)

Graph 1 — Latency vs Offered Load:

- **Bare Metal:** Latency stays extremely low (~2ms) across all measured load levels, never approaching the 500ms SLO. The curve is essentially flat, indicating the system is nowhere near saturation.
- **HW VM:** Very similar to bare metal, staying around 2-3ms up to 22,000 req/s before latency begins rising sharply, crossing 500ms around 22,000–25,000 req/s. The hardware virtualization overhead is minimal.
- **TCG VM:** Latency starts higher (~90-100ms) even at low loads due to software emulation overhead. It crosses the 500ms SLO between 1,250 and 1,500 req/s and explodes to over 11,000ms at 2,500 req/s, showing very early saturation.

Graph 2 — Throughput vs Offered Load:

- **Bare Metal & HW VM:** Both track the ideal diagonal closely, meaning the system handles almost every request it receives without dropping any.
- **TCG VM:** Tracks the ideal line up to about 1,250 req/s, then throughput flattens and caps around 22,000 req/s regardless of how much load is offered — the system is saturated and dropping requests.

Graph 3 — Latency vs Throughput & Max Points:

- **Bare Metal:** Latency stays flat and low across the entire throughput range. The max point is at the highest measured throughput (~52,000 req/s) while still under 500ms — meaning bare metal can sustain very high throughput within the SLO.
- **HW VM:** Similar shape to bare metal but saturates earlier. The max point is around **21,956 req/s** — the highest throughput achievable while keeping 99th percentile latency below 500ms.
- **TCG VM:** The curve rises steeply very early. The max point is around **1,248 req/s** — beyond this, latency exceeds 500ms and the system is overloaded.

The max point signifies the **practical capacity limit** of each technology — it is the maximum useful throughput you can get while still providing acceptable service to 99% of users.

ii) Which technology provides the best performance? Why does each technology provide each level of performance? (5 Marks)

Bare Metal performs best because there is no virtualization layer at all. The application runs directly on the physical hardware, with no overhead from hypervisors, virtual network stacks, or emulated devices. Every CPU cycle and network packet goes directly to the application.

HW VM performs close to bare metal because modern CPUs have built-in hardware virtualization support (Intel VT-x). This means privileged instructions from the VM are handled directly by the CPU hardware rather than being intercepted by software. The network IO path still goes through a software virtual switch (vswtch), which introduces some overhead — explaining the slightly higher latency and lower saturation point compared to bare metal.

TCG VM performs worst because it uses software emulation — every CPU instruction the VM executes must be translated and emulated in software by QEMU's Tiny Code Generator. This is extremely expensive. On top of that, all IO operations are also emulated in software. The result is massive overhead that explains why latency is already high at low loads and the system saturates at just ~1,248 req/s compared to ~22,000 req/s for HW VM.